Project on

AI POWERED AUTONOMOUS COCONUT HARVESTING SYSTEM

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2025-2026

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INTRODUCTION

Coconut farming is one of the major agricultural activities in India, especially in Kerala and Karnataka.
But harvesting coconuts is still done manually by skilled climbers, which is:

- Risky (fall injuries, accidents)
- Labor-intensive (fewer people willing to climb)
- Time-consuming and inconsistent.

Our project introduces an AI-powered hybrid system — combining drones for smart coconut detection and robots for adaptive harvesting — to make the process safe, efficient, and scalable."

EXISTING SYSTEM

S.No.	Patent / Publication Name	Key Features	Estimated Overlap	Novelty Status	Source Link
1	CN217064635U – Automatic Coconut Picking Device	Eight-wheel climbing robot with OpenMV vision and mechanical claw	60% – Vision-guided climbing and harvesting; lacks hybrid AI/manual mode	Strong Functional Similarity	Google Patents – CN217064635U
2	Cocobot – Coconut/Areca Nut Harvester (ICAR-CCARI & Goa University)	Remote-controlled coconut harvester with camera and robotic arms	55% – Semi-autonomous, lacks integrated AI or dynamic climbing adaptability	Conceptually Similar; Technically Simpler	Krishi Jagran News
3	ARCH-1 – Advanced Remote-Controlled Coconut Harvester	Dual robotic arms, live video, fixed mechanical climber	45% – Remote-controlled system; no adaptive AI or environmental sensing	Proof-of-Concept Overlap	Indian Journal of Science & Tech

PROBLEM STATEMENT

Coconut harvesting is a labor-intensive process that depends on skilled workers, making it time-consuming and inefficient. Farmers often struggle to accurately identify the maturity of coconuts, which leads to improper harvesting and reduced yield quality. Additionally, there is a lack of smart systems that provide real-time monitoring and decision support, especially in rural areas where internet access is limited. This creates a need for an intelligent and user-friendly solution to assist farmers in efficient coconut harvesting

“The problem is clear — coconut harvesting today is unsafe and inefficient… but we are building the solution: an AI-powered hybrid system that makes harvesting safer, smarter, and scalable.”

OBJECTIVES

- To develop a system for detecting coconuts using image processing techniques
- To classify coconuts based on maturity using features like color, size, and shape
- To design a mobile application for displaying results and recommendations
- To provide real-time data for better decision-making an to enable offline functionality for rural usage
- To integrate location tracking (latitude & longitude) for farm monitoring
- To support future integration with robots or drones for automated harvesting
- To develop a cost-effective and easy-to-use solution or farmers

PROPOSED SYSTEM

The proposed system works by capturing images of coconut trees using a mobile device or drone and processing them through a trained deep learning model to detect and classify coconuts based on their maturity. The system analyzes features such as color, size, and shape to determine whether the coconut is ready for harvesting.

A trained Al classification model is then applied to differentiate between:

- Mature coconuts (ready for harvesting)
- Immature coconuts (not ready for harvesting)

The processed results are sent to a mobile application, where users can view real-time information about coconut status along with harvesting recommendations. The system also incorporates location tracking to organize farm data and is designed to function in offline conditions. This approach enables efficient monitoring and supports timely and accurate harvesting decisions

BASE PAPERS

Sl. no	TITLE OF THE PAPER	AUTHOR	YEAR
01	PickPerfect: Coconut Maturity Detector and Harvester Using Deep Learning	Dr. Amitha I C, Fathima Fiza C P, Aswin Ratheesh D K, Rena Haris V P, K Muhsin	2025

Key Points:

- Uses CNN-based computer vision model to classify coconut maturity stages it analyzes features like color, texture, and size for accurate detection
- Includes a mechanical harvesting system with adjustable arms for picking coconuts and Improves harvesting efficiency by selecting only mature coconuts.

Limitations: Requires controlled conditions for accurate image capture and Mechanical system needs manual support and is not fully automated

Gaps: Needs real-time field deployment and testing and it requires integration with mobile apps and loT systems and Improvement needed for fully automated harvesting

BASE PAPERS

Sl. no	TITLE OF THE PAPER	AUTHOR	YEAR
02	Computer-Vision-Based Detection and Monitoring System for Mature Coconut Fruits with a Web Dashboard Visualization Platform	Samfford S. Cabaluna, Maria Fe P. Bahinting, Leah A. Alindayo	2024

Key Points:

- Introduces a YOLOv8-based system to detect and classify mature vs. premature coconuts with 99.5% precision and recall.
- Designed as a monitoring and decision-support tool, not just detection, offering better usability for farmers.

Limitations: The system was tested only in the laboratory with close-range datasets, misclassifies coconuts in some cases, and is not yet deployed in real-time or field environments.

Gaps: Future improvements require larger on-tree datasets, drone-based or distant-angle detection, real-time deployment, and integration with robotic harvesters or mobile apps to make the system fully practical.